

**GUIDELINES FOR THE
IDENTIFICATION OF MUTATIONS
USING HIGH-THROUGHPUT
SEQUENCING:
The Case of Banana and Beneficial Microbes**

**The Joint FAO/IAEA Centre of Nuclear Techniques in Food and
Agriculture**

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FOREWORD

This document represents a strategic effort to streamline and standardize the application of high-throughput sequencing in the dual context of crop improvement and biological disease control. Developed as a comprehensive technical framework, these guidelines are designed to facilitate the identification of genomic variation induced by gamma irradiation in both banana (*Musa spp.*) and their associated beneficial microorganisms.

As global banana production faces intensifying pressure from evolving pathogens and climate instability, the integration of mutation breeding and advanced genomics has become essential. This guide provides a roadmap for navigating the key stages of modern sequencing workflows, from sample preparation and DNA extraction to the library preparation to the selection of appropriate sequencing platforms tailored to specific research objectives.

By bridging the gap between induced mutagenesis breeding and genomics, these guidelines aim to support two main objectives:

- Enhancing breeding pipelines: Accelerating the identification of elite banana mutants with improved agronomic and/or disease resistance traits, while strengthening understanding of plant defense mechanisms and host-pathogen interactions.
- Optimizing biological control strategies: Supporting the genomic characterization of beneficial microorganisms to improve their efficacy within integrated pest management programs, specifically for the biological control and growth promotion in disease infected fields.

Ultimately, this document is intended to serve as a practical resource for researchers and breeders in IAEA and FAO Member States, providing guidance to ensure that sequencing projects are well designed and that resulting genomic data can be effectively used for downstream analyses in subsequent products.

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I. Introduction

The process of generating annotated genome sequences to support mutation breeding follows a series of common steps, regardless of the sequencing technology used. However, the specific tools and analytical approaches employed to performed specific steps may vary depending on the platform and biological system under study. Once a clear biological question has been defined, sequencing projects can be broadly divided in three phases i) pre-sequencing phase, ii) sequencing, and iii) post-sequencing phase (**Figure 1**)

The **pre-sequencing** phase includes the identification of appropriate samples and controls; selection of a suitable sequencing service provider, choice of sequencing technology and determination of whether a reference genome is available. These steps are critical for ensuring that the resulting data are suitable for detecting mutations associated with phenotypes of interest. This document primarily addresses the pre-sequencing phase, , including guidelines of sample collection, DNA extraction and the selection of sequencing platforms.

The **sequencing** phase involves library preparation and generation of raw sequencing data using the selected technology. General principles of sequencing technologies are introduced in this document to support informed decision-making.

The **post sequencing** activities include, retrieving the raw sequencing data, quality control, assembling the genome(s) either de novo or using a reference; evaluating the quality of the assembled genome; genome annotation and downstream analysis to identify candidate genomic regions associated with the observed mutant phenotypes. Detailed bioinformatic workflows for these steps will be provided in in the subsequent documents (Products 2 and 3).

II. Pre-sequencing phase

The pre-sequencing phase includes the identification of appropriate samples and controls, selection of a suitable sequencing service provider, choice of sequencing technology, and confirmation of the availability of a reference genome. These steps are critical because errors or biases introduced at this stage cannot be corrected during downstream bioinformatic analyses.

All aspects of the pre-sequencing phase are addressed in this document, including sample collection, tissue preservation, DNA extraction, quality assessment, metadata recording, and shipping of material for sequencing.

1. General guidelines for sample collection and tissue preservation for sequencing projects.

Where to sequence the samples

Identify a suitable sequencing service provider, this is critical as it needs to be determined ahead of time. The distance and duration of sample transport can strongly influence the quality of plant tissue and the integrity of extracted DNA or RNA. **Table 1** lists examples of sequencing providers located in regions where banana production and research is highly relevant. This information is intended to facilitate access to sequencing services near collaborating laboratories.

The following locations have been considered, based on their active research in bananas; South and Central America (e.g. Ecuador, Colombia, Guatemala, Costa Rica, Brazil, and Honduras), Africa (e.g. Uganda, Kenya, Rwanda, Burundi, Tanzania, Democratic Republic of Congo (DRC), west Africa, Nigeria, Mozambique, South Africa, Ethiopia), Asia (e.g. South East Asia - Philippines, Indonesia, Vietnam, Thailand, Malaysia, and India; China and Taiwan).

If samples are transported across national borders, or between regions within the same country with different regulatory frameworks, all required permits such as phytosanitary certificates and/or material transfer agreements must be obtained in advance to prevent any delays due to quarantine or missing documentation. Researchers are strongly encouraged to become familiar with all the needed documents, agreements and protocols prior to project initiation.

When shipping biological material for sequencing, appropriate temperature control must be maintained throughout transport. RNA samples and fresh tissue should be transported on dry ice, while DNA samples are typically shipped on ice or cold packs, however, high-quality purified DNA can also be transported at ambient temperature for short periods, provided it is properly sealed and protected from degradation. These conditions should be ensured from the point of departure to final delivery in order to preserve nucleic acid integrity and maximize the likelihood of obtaining high-molecular-weight DNA or intact RNA.

Metadata and vouchers

Metadata collection should be as complete as possible for each genome to be sequenced. If working with wild relatives, for instance in cases where a reference genome will be generated, metadata collection should follow international standards as recommended by the Earth BioGenome Project (Lawniczak et al., 2025). For breeding materials, following the MIAPPE 1.1 standard (Papoutsoglou *et al.*, 2020) is recommended.

Vouchers are recommended for all samples and should allow an unambiguous association between physical material, associated metadata and resulting data outputs.

Sample collection

Researchers are advised to consult their sequencing service providers for specific requirements related to sample collection, especially for amount of tissue required preferred preservation methods for the chosen sequencing technology.. Most sequencing facilities prefer to receive cold-preserved tissue rather than receiving extracted DNA. However, in some cases, researchers may choose to extract the DNA to keep costs low.

2. DNA extraction and quality assessment

If DNA will be extracted prior to submission for sequencing, the following are recommendations apply for plant tissue collection:

1. Collect tissue from 10-12 days old seedlings.

2. Dark treatment for 48-72 hours prior to sampling, while maintaining hydration, can improve DNA quality.
3. If the objective includes detection of copy number variation and endoreplication is a concern (excluding clonally propagated mutants), replicate sampling from different tissue types may be considered., however, this is not very common when working with banana cultivars.
4. Both mutant and non-mutant (control/wild type) samples should be included whenever available, rather than relying solely on a reference genome.
5. At least 100 mg of tissue per 1 gigabase (Gb) of genome size should be collected.
6. High molecular weight DNA extraction kit or protocols are recommended.

Platform-specific guidelines for DNA extraction and QC are available for PacBio: [DNA-extraction-for-PacBioHiFi-Sequencing](#); ONT: [DNA-extraction-for-ONT-sequencing](#); and Illumina: [Library-preparation-considerations-Illumina](#).

If specialized protocols are not available, widely used CTAB based extraction methods (Doyle and Doyle, 1987) can be applied. A CTAB-based method optimized for banana is provided below:

CTAB-method with 0.3% v/v β -mercaptoethanol – Taken from Koh et al., 2021 for extractions from *Musa textilis*.

The protocol (Protocol 3) was based on Healey et al. (2014) with minor modifications.

1. One hundred milligram of tissue sample was ground into a fine powder using liquid nitrogen, and 800 μ L of CTAB DNA extraction buffer (100mM Tris-HCl pH 7.5, 25 mM EDTA pH 8.0, 1.5 M NaCl, 2% w/v CTAB) supplemented with 0.3% (v/v) β -mercaptoethanol was added
2. The mixture was further homogenized and was transferred to a 1.5 mL tube
3. Tubes were incubated at 65°C for 30 min with constant inversion every 10 min. Solid debris were separated from the liquid portion through centrifugation (15000 $\times$ g for 5 min at RT)
4. The resulting supernatant (600 μ L) was transferred to a new 1.5 mL tube and an equal volume (600 μ L) of 24:1 chloroform:isoamyl alcohol was added
5. The mixture was constantly inverted for 5 min to fully mix the solution and the aqueous phase was separated by centrifugation (15000 $\times$ g for 5 min at 4°C)
6. Five hundred microliters of the recovered aqueous phase was transferred to a new 1.5 mL tube and RNA was digested with the addition of 0.5 μ L of 10,000 ppm RNase A
7. The mixture was incubated at 37°C in a dry bath for 15 min with inversion every 5 min
8. Chloroform:isoamyl alcohol extraction was repeated, and 400 μ L of the resulting supernatant was divided among two 1.5 mL tubes
9. DNA was precipitated by addition of $\frac{1}{2}$ volume (100 μ L) of 5 M NaCl and 3 volumes (600 μ L) of ice cold 95% ethanol and incubation at -20°C for 30 min
10. Tubes were centrifuged at 15000 $\times$ g for 5 min at 4°C to pellet the DNA precipitates
11. Pellets were washed twice with 600 μ L of 70% ethanol (centrifuged at 15000 $\times$ g for 5 min at 4°C) and were resuspended in 15 μ L of nuclease-free water
12. The resuspended DNA (15 μ L) from each tube were pooled together to a final volume of 30 μ L
13. Run the extracted DNA on a 0.7% agarose gel for confirmation of integrity.

Additionally, there are commercially available kits commonly accepted such as the DNeasy Plant Kits from Qiagen ([Plant-DNeasy](#)) that has been used for banana leaf tissue. Regardless of the extraction method used, two practices are strongly recommended; first is tissue disruption using a mortar, a pestle and liquid nitrogen and second using wide-bore pipette tips throughout the protocol to minimize excessive DNA shearing.

To extract DNA from microorganisms, for bacterial samples, wild-type and mutant strains should be re-streaked and isolated from single colonies to avoid mixed populations. For Illumina sequencing, DNA extracted from cultures in logarithmic or early stationary growth phase is generally sufficient, provided DNA purity is high. For long read sequencing (PacBio/ONT), DNA purity and length are more critical than growth phase..

For fungi, collection of single-spore isolates or a clean hyphal tip cultures ecommended to minimize heterokaryosis. Commercial kits optimized for high-molecular-weight DNA extraction from bacteria and fungi are available, including the Monarch HMW DNA Extraction Kit from New England Biolabs ([Monarch-HMW-DNA-Extraction-Kit-for-Tissue.](#)) and the Quick-DNA HMW MagBead Kit from Zymo Research ([Quick-DNA-HMW-Magbead-Kit](#)), both of which have been reported to yield DNA suitable for long-read sequencing. (Gand et al., 2023).

Ideally, extracted DNA should exceed 50 kb for PacBio and Illumina and 100 kb for ONT) Optical density ratios should be approximately 1.8 for 260/280 and ≥ 2.0 for 260/230 when measured using a spectrophotometer.

Assessment of DNA integrity using a Bioanalyzer, Tape station or a Femto Pulse system is recommended. Standard agarose gel electrophoresis has limited resolution above 20 Kb and may not reveal degradation that can bias library preparations. However, if advanced instruments are not available, DNA can be run on a 0.7% agarose gel to confirm the presence of high molecular weight DNA (**Figure 2**). Final quality control is typically performed by the sequencing facility, which may provide feedback or request re-extraction if quality thresholds are not met.

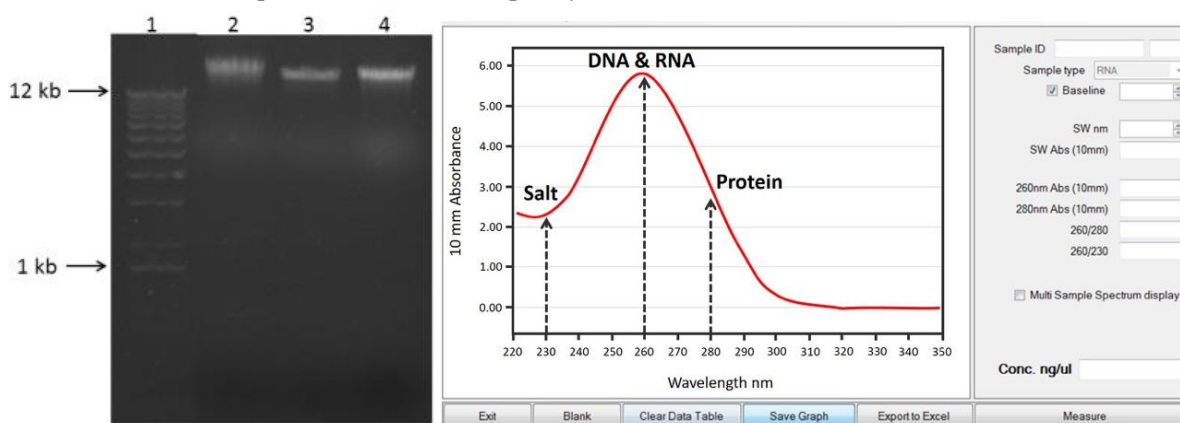


Figure 1. Quality of extracted DNA. Left: an example figure showing high molecular weight DNA run on a 0.7% agarose gel and stained with SYBR Safe DNA gel stain. As shown in the image the bands should barely migrate and stay at above the 12 kb marker band, do not show any indications of smear. Right, if a nanodrop is available, a similar profile curve should be obtained, 260/280 values near 1.8 and 260/230 near 2.0 are indicators of good quality DNA. Image was taken from Healey et al., 2014.

Table 1. High throughput sequencing providers in banana grow and research regions

Vendor of Services	Country	Contact	Offered Technologies	Service Region
Novogene	China	Novogene inquiry_us@novogene.com	PacBio, Illumina, Oxford Nanopore	Global
CD Genomics	US	CD_Genomics contact@cd-genomics.com	PacBio, Illumina, Oxford Nanopore	Global
BGI Genomics	China	BGI info@genomics.cn	PacBio, Illumina, Oxford Nanopore	Global
Core Lab Genomics - Monterrey Tecnológico	México	Core_Lab_Genomics	Illumina NovaSeq 6000	Central América
BGI Americas	Brazil	BGI Americas	BGI Genomics	South América
GenCore – Universidad de los Andes (Bogotá)	Colombia	GenCore Los Andes gencore@uniandes.edu.co	Oxford Nanopore	South América
Inqaba biotec	South Africa -Pretoria and Cape Town	Ingaba Biotec info@inqababiotec.co.za	PacBio, Illumina	Southern and South-east Africa
Krisp	South Africa	Krisp deoliveira@ukzn.ac.za	Illumina, Oxford Nanopore	South Africa
Kibs	Kenya - Nairobi	Kibs info@kibs.co.ke	Unknown	Kenya
Philippine Genome Center	Philippines	PhilippineGenome Center dnasequencing@pgc.up.edu.ph	Illumina, PacBio, Ion Torrent	Philippines
Biotools	Taiwan	Biotools ngs@toolsbiotech.com	PacBio	Taiwan
Annoroad	China - Beijing	Annoroad zj-annoroad@annoroad.com	PacBio, Illumina	China
GrandOmics	China - Beijing	GrandOmics	PacBio, Illumina	China, Global
LOBI	Vietnam	Lobi info@lobi.vn	PaBio, Illumina	Vietnam
Genotypic	India - Bangalore	Genotypic genomics@genotypic.co.in	PaBio, Illumina, Oxford Nanopore	India
SciGenom	India - Kerala	SciGenom	Illumina	India, US

3. Availability of a reference genome and guidelines for generating one

The generation of a high-quality reference genome for banana (*Musa* spp.) requires approaches that account for interspecific hybridization, polyploidy, and a repeat-rich genome. Several reference-grade assemblies are currently available (**Table 2**); however, Additional genomes are needed to represent rare genome types and wild relatives.

For the generation of a new reference genome, the recommended approach is to use PacBio HiFi sequencing as the primary data source, targeting sufficient coverage to minimize haplotype or sub-genome collapse (typically $\geq 50\times$ for hybrid and triploid genomes). This may be complemented by high-throughput chromosome conformation capture (Hi-C) technology (Wang et al., 2022) to enable chromosome-scale scaffolding and structural validation. **Table 3** summarizes some of the technical requirements for generating a reference genome.

Chromosome-level assemblies are generally sufficient for most biological, breeding, and comparative genomics applications. In contrast, telomere-to-telomere (T2T) assemblies should be considered mainly for diploid progenitor species or for studies focused on centromeres and repetitive genome architecture. **Table 4** compares the main characteristics and intended uses of chromosome level and T2T assemblies.

Chromosome-level assemblies are suitable for most applications relevant to mutation discovery, including gene identification, QTL mapping, and comparative genomics. T2T assemblies are primarily of value for specialized studies of centromeres, telomeres, and highly repetitive regions, and are not required for routine mutation discovery workflows.

When developing a reference genome, emphasis should be placed on structural correctness, sub-genome awareness, and robustness, as deficiencies in the reference genome directly propagate errors into downstream read mapping, variant detection, and population genomic analyses.

The procedures for sample collection and DNA extraction for reference genome sequencing should follow the same guidelines outlined in the pre-sequencing phase of this document.

Where a suitable reference genome is already available, it should be used in preference to generating a new assembly. Selection of an appropriate reference genome should consider genome group (e.g. AA, BB, AAA), ploidy level, and phylogenetic proximity to the study material.

If only a distantly related reference genome is available, it may still be used with caution. In such cases, researchers should be aware of potential reference bias and reduced sensitivity for detecting structural variants and copy number changes.

Table 2. Banana genomes available, suitable to be used as reference genomes. Genomes have been identified for different *Musa* species (Arshad et al., 2025). Assemblies have been listed based on a BUSCO score of 95 or above. Only genomes with available data to be downloaded are listed and links are provided

Species/Cultivar	Genome group	Chr. No.	Technology	Size (Mb)	Contig N50 (Mb)	BUSCO (%)	Version	Link to data	References
<i>M. acuminata</i> ssp. <i>malaccensis</i>	AA	11	ONT	468.82	32.00	98.80	V4	Genome at Genoscope Genome at Banana Genome Hub Genome at ENA	Belser et al., 2021
<i>M. acuminata</i> ssp. <i>malaccensis</i>	AA	22	PacBio, ONT, Hi-C	940	43.10	98.57	V1	Genome at NCBI (PacBio) Genome at NGDC Genome at NCBI (Illumina)	Liu et al., 2023
<i>M. acuminata</i> ssp. <i>zebrina</i>	AA	11	Illumina, ONT	535.90	8.30	98.77	V2	Genome at NCBI	Martin et al., 2025
<i>M. acuminata</i> ssp. <i>banksii</i>	AA	11	Illumina, ONT	486.80	1.50	96.00	V1	Genome at NCBI	Martin et al., 2025
<i>M. acuminata</i> cv. Pisang Madu	AA	22	PacBio	955.70	28.80	98.80	V1	Genome at NCBI Genome at NCBI	Martin et al., 2025
<i>M. acuminata</i> cv. Baxijiao	AAA	33	PacBio, ONT, Hi-C	1,424	44.80	96.40	V1	Genome at NCBI Genome at NGDC	Huang et al., 2023
<i>M. acuminata</i> cv. Gros Michel	AAA	33	PacBio	1,331	1.30	96.90	V1	Genome at NCBI	Li et al., 2024
<i>M. schizocarpa</i>	S	11	Illumina, ONT, Bionano	525.30	6.50	97.60	V1	Genome at ENA	Belser et al., 2018
<i>M. schizocarpa</i>	S	11	Illumina, ONT	509.80	20.70	98.70	V2	Genome at NCBI	Martin et al., 2025
<i>M. ornata</i>	BB	11	ONT	477.18	12.88	98.08	V1	Genome at NGDC	Xiao et al., 2024
<i>M. velutina</i>	AA	11	ONT	496.23	18.18	98.51	V2	Genome at NGDC	Xiao et al., 2024
<i>M. balbisiana</i>	BB	11	ONT	492.80	2.00	96.90	V1	Genome at CNGBdb Genome at Banana Genome Hub Genome at NCBI	Wang et al., 2019
<i>M. troglodytarum</i> cv. Karat	AT	10	Illumina, PacBio	603.60	5.10	97.70	V1	Genome at NCBI	Li et al., 2022
<i>M. textilis</i> cv. Abaca	TT	10	ONT	613	3.50	96.90	V1	Genome at NGDC	Zhou et al., 2024
<i>M. textilis</i>	TT	10	Illumina, ONT	531.10	4.00	95.80	V2	Genome at NCBI	Martin et al., 2025

Table 3. Technical requirements for reference genomes

Goal	PacBio Hi-Fi (backbone)	Ultra-long ONT	Hi-C (chromosome scaffolding)	Illumina Short reads	Expected outcome
Chromosome-level (standard “reference-grade”) - diploid	30–45x	5-15x Optional for repeats or structural resolution	100-300M read pairs per Gb genome	Optional 20–40x (for QC and k-mer validation)	Chromosome-scale pseudomolecules (may still contain gaps), near-complete gene space, structural correctness, optional alternate haplotigs partial phasing depending on heterozygosity and pipeline.
Chromosome-level - hybrid/polyploid	50–70x	Optional 10–30x	200–600M read pairs (higher to compensate for complexity/noise)	Optional 30–60×	Chromosome-scale pseudomolecules with reduced homeolog collapse; improved sub-genome and haplotype separation; more reliable Hi-C scaffolding in repetitive and hybrid genomes; suitable for structural variation, CNV, and sub-genome-specific gene annotation
T2T -diploids	40–60x	≥ 30x (centromeres, rDNA arrays, telomeres, satellite repeats)	100–300M read pairs (optional for anchoring and validation)	Optional 30–50x	Gapless chromosomes, including complex repeats, centromeres and telomeres
T2T - hybrids/polyploid	≥ 60x	≥40x	400–800M read pairs	Optional 30–60×	Technically feasible for some genomes but extremely fragile; high risk of homeolog mis-joins, false merges, and incorrect centromere assignment.

Table 4. Comparison of chromosome level and T2T reference genomes

Attributes	Assembly Type	
	Chromosome-level	T2T (Telomere-to-Telomere)
Definition of the assembly type	Chromosomes represented as pseudomolecules with gaps	Each chromosome is a single, gapless sequence
Continuity	Scaffolded; contigs joined with Ns	Fully continuous (no Ns)
Telomeres	Often partial or missing	Fully assembled on both ends
Centromeres	Usually collapsed or masked	Fully resolved centromeric repeats
Pericentromeric repeats	Fragmented or missing	Fully represented
rDNA arrays	Broken into multiple contigs	Continuous arrays
Gene space	Near complete	Complete
Structural accuracy	High at chromosome scale	Highest possible (true structure)
Base accuracy	Very high	Very high
Repeat representation	Incomplete	Near-complete
Hi-C contact map quality	Clean, but gaps present	Clean <i>and</i> continuous
Genome size estimate	Often underestimated	Accurate (true size)
Data typically required	PacBio HiFi + Hi-C	HiFi + ultra-long ONT + Hi-C
Assembly complexity	Moderate	Very high
Time & curation	Weeks	Months
Best use	Gene discovery and annotation, QTL mapping, GWAS, comparative genomics, synteny, breeding and marker development, pangenomes, regulatory regions near genes	Centromere structure and evolution, repeat expansion/contraction, rDNA copy number and organization, chromosome architecture modeling, true genome size determination, structural variation in repeats, telomere dynamics

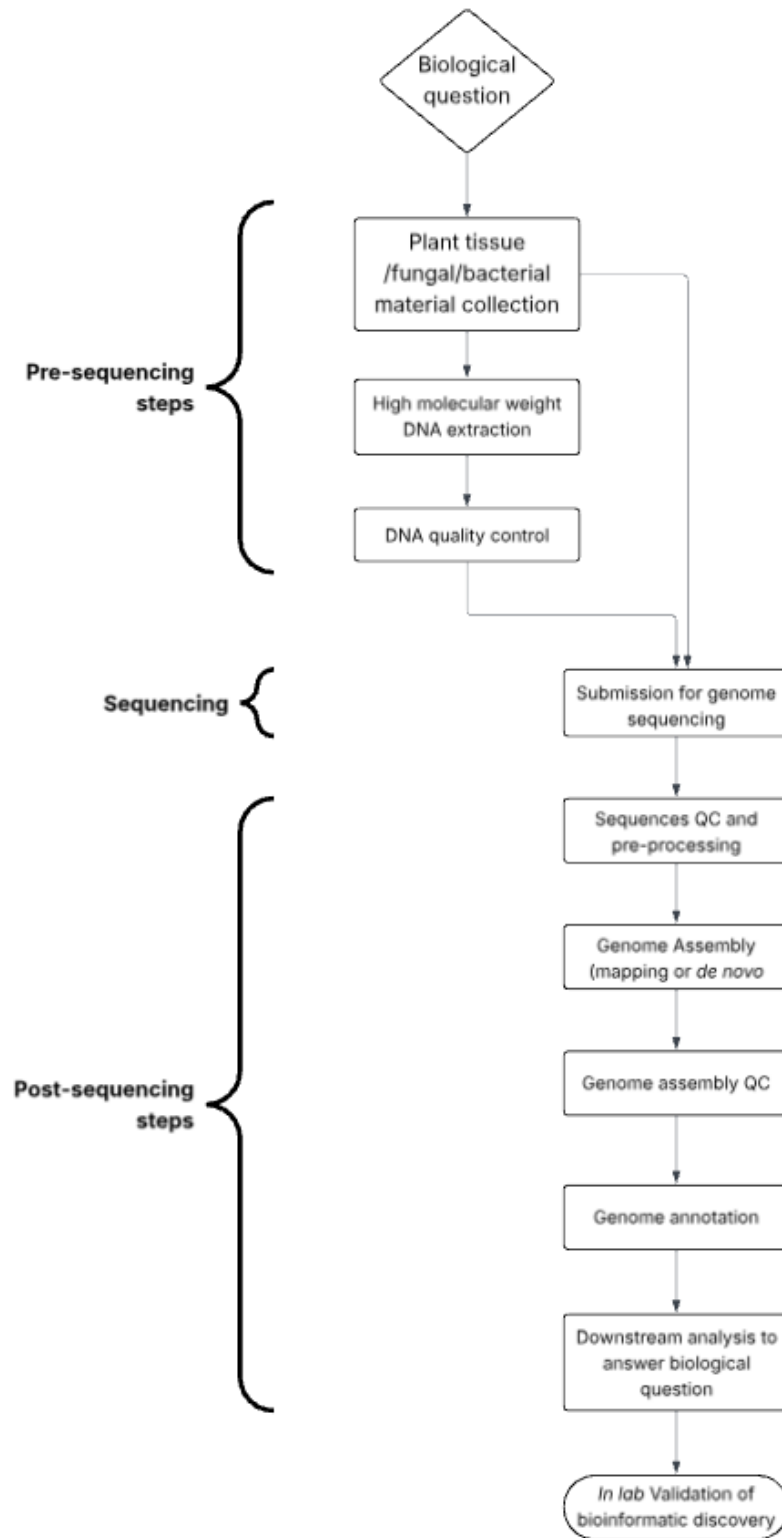


Figure 2. Project workflow depicting pre- and post- sequencing phases.

III. Sequencing phase:

1. Sequencing technologies

Currently, two major categories of high throughput sequencing technologies are widely used, based on read length. Long-read sequencing such as PacBio and Oxford Nanopore (ONT) technologies and short read sequencing such as Illumina.

PacBio sequencing - is based on single-molecule real-time (SMRT) technology, in which a DNA polymerase is observed within zero-mode waveguides, enabling long reads ranging from 10-25 kb. The consensus sequencing approach produces high fidelity (HiFi) long reads exceeding 99.9% accuracies, making this technology well suited for de novo genome assembly, structural variant detection, and haplotype phasing (Eid et al., 2009; Wenger et al., 2019).

Oxford Nanopore sequencing - detects changes in ionic current as single DNA or RNA molecules pass through a nanopore, enabling generation of ultra-long reads (often exceeding 100 kb, potentially megabases), real-time sequencing, as well as potential detection of base modifications. Although raw error rates are higher, continued improvements in chemistry and base-calling algorithms, combined with error correction using complementary technologies such as Illumina or PacBio, have substantially increased data quality (Jain et al., 2016; Deamer et al., 2016).

Illumina sequencing - based on sequencing-by-synthesis using reversible terminator nucleotides, produces short reads, typically 100–300 bp in length, with very high base accuracy. This technology is particularly suitable for SNP detection transcriptome profiling, and high-depth re-sequencing (Bentley et al., 2008; Goodwin et al., 2016). Illumina sequencing can be single-end (SE) or pair-end (PE). In SE sequencing the instrument reads DNA fragments from only one side, producing a single FASTQ file (R1). This approach is simpler and cheaper but provides less positional information, which can limit mapping accuracy in repetitive or complex regions. In PE sequencing reads both ends of the same fragment, generating two FASTQ files (R1 and R2) with a known insert size between them. This extra information greatly improves alignment, variant detection, and genome assembly, especially in repeat-rich genomes, at the cost of increased sequencing effort and expense.

3. Identify best sequencing technologies for each project, based on the underlying scientific questions

Gamma irradiation is widely used in breeding programs to generate genetic variation in crops (FAO-IAEA, 2017). Mutations induced by gamma radiation are commonly represented by insertions, deletions and copy number changes, depending on the applied dose. Once mutant phenotypes of interest have been identified through established screening pipelines, the next step is to determine the underlying genomic changes responsible for these traits..

The decisions of how to generate that information will depend on the resources available such as technology, reference genome, and ploidy of the target genotype. Two main scientific goals are addressed in this document. The first one is to identify genomic regions in a mutant banana plant genotype that rendered a phenotype of interest. The second one is to identify the genomic regions of

mutated microorganism that through mutation technologies has rendered into a beneficial organism for a banana plant, for instance for the control of a disease.

Bananas - Mutation breeding has been used to generate multiple banana varieties with improved resistance to disease. Several resistant mutants (e.g. DPM25, Highgate, Novaria, and Lakatan BBTV) to *Fusarium oxysporum* f.sp. *cubense* have been reported following gamma irradiation (Ahloowalia et al., 2004; Hwang et al., 2007; Jankowicz-Cieslak et al., 2012; Datta et al., 2018; Mintoff et al., 2021; Adero et al., 2023).

When designing sequencing-based studies to identify genomic changes underlying mutant phenotypes in banana, several factors should be considered. These include the availability of a non-mutant control line, the availability and suitability of a reference genome, and the genome composition and ploidy level of the cultivar under study. A suggested decision tree for selecting the best sequencing strategies based on these factors is shown in **Figure 3**. The main decision points are:

- (i) whether a non-mutant control is available
- (ii) whether a closely related reference genome exists
- (iii) whether only a distantly related reference genome is available.

Given the types of mutations typically induced by gamma irradiation, long read sequencing technologies are generally the most suitable for mutation discovery in banana, particularly for the detection of large insertions, deletions, and copy number variants. Among these technologies, PacBio HiFi sequencing is recommended as a primary option due its continuous read length of 10-25,000 bp and high base accuracy of 99.9% (0.1% error) (Wenger et al., 2019).

However, in cases where mutation parameters are well optimized and large populations are being sequenced, especially in early phases of screening, short-read sequencing at low coverage may be used for preliminary detection of large genomic changes, as previously demonstrated using Illumina data (Datta et al., 2018). **Table 5** summarizes indicative sequencing depths and recommended applications for different sequencing technologies in banana resequencing studies.

Banana genomes comprise multiple genome types (A, B, S, and T) and ploidy levels (e.g. AA, AB, AAA, AAB, ABB, and more complex combinations BBB, TT, AT, AAAB, AABB, AB BB, AS, AAT and BBT) (Rekha et al., 2016; Zhou et al., 2017; Wang, et al. 2019; Zhou, et al. 2024). Ideally, a reference genome for each genome type would be available., However, the generation of new reference genomes is resource-intensive and may not be feasible for all projects.. Currently there are several reference level genome assemblies available for utilization and they were listed on **Table 2**. Guidelines for generating additional reference genome are presented in **Section 2**. Where a suitable reference genome is unavailable and a non-mutant control (wild type) cannot be obtained, existing reference genomes from related genome groups may be used with caution. In such cases, researchers should be aware of potential reference bias and reduced sensitivity for detecting structural variants and copy number changes. (Chen et al., 2021; Akopyan et al., 2025).

Table 5. Recommended target depth of re-sequencing by technology

Sequencing technology	Target depth	Common applications
PacBio HiFi	30X	Assembly, structural variants (SV), deletions and copy number variants (CNV)
PacBio HiFi + ONT	HiFi: 30–40X + ONT: 40–60X	Very long repeats, big rearrangements, or want near-T2T contiguity
PacBio HiFi	50X	Copy-number/dosage especially in triploids
Illumina	100X	

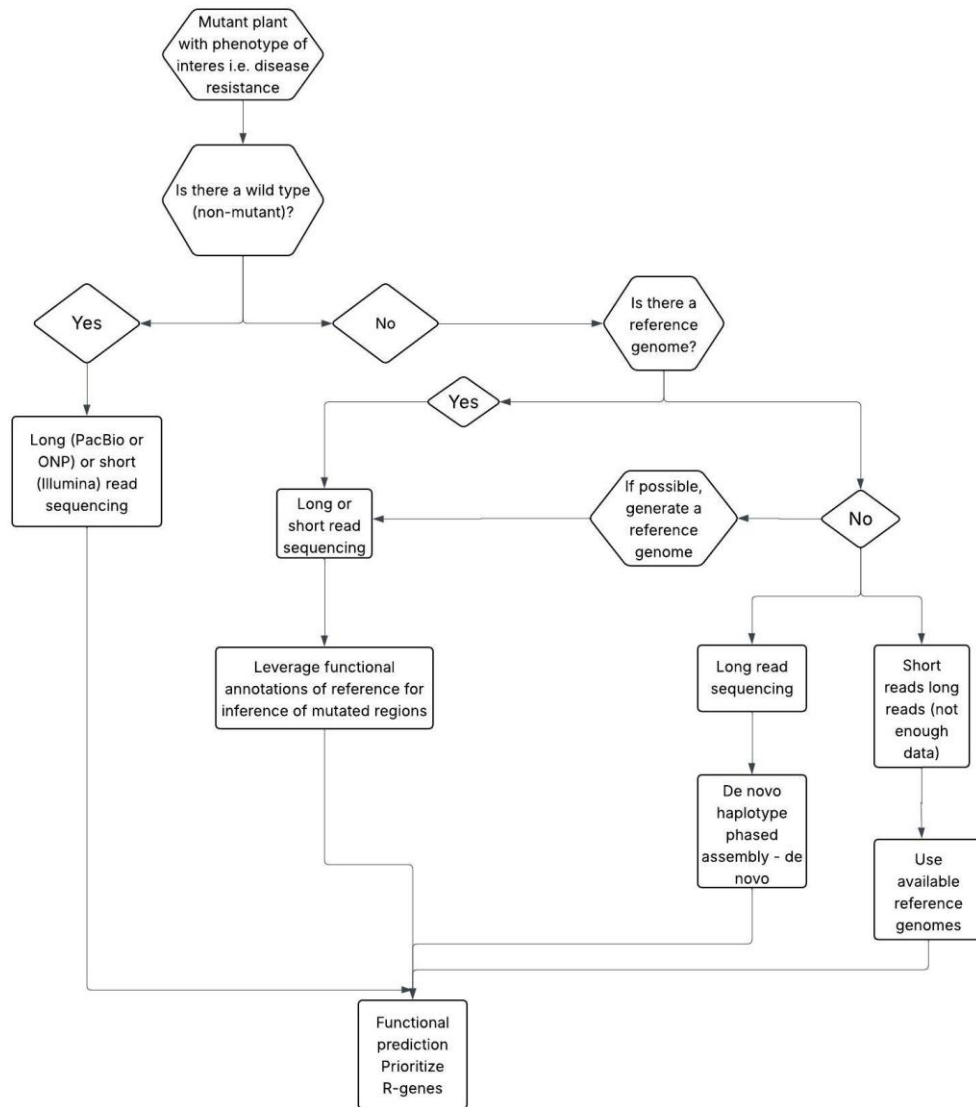


Figure 3. Decision tree for selecting sequencing approach for banana projects

Beneficial microorganisms – Beneficial microorganisms have been identified as effective biological control agents banana diseases such as Fusarium wilt. . For instance, *Streptomyces* sp. XY006 (Shan et al., 2017; Shan et al., 2018) has shown to be an effective inhibitor of pathogenic fungi such as *Fusarium oxysporum* f. sp. *cubense* (Foc), to date the most devastating disease for bananas. Other microorganisms such as *Trichoderma* (Tao et al., 2023), *Pseudomonas* (Lv et al., 2023), *Bacillus* (Xiang et al., 2023) and *Streptomyces* (Zhu et al., 2023) have also been reported to be effective in controlling Foc, and therefore could be used as effective biological controls.

Generating mutations through gamma irradiation is a common approach to generate genetic changes in microorganisms with enhanced antagonistic properties such as increased production of antimicrobial metabolites, improved host colonization and greater ecological stability (Mrimajlessi et al., 2018; Palmieri et al., 2022; Rostami et al., 2024).

The identification of the underlying genomic regions in mutant microorganisms will always rely on the re-sequencing of the wild type (or non-mutant control) along the mutant microorganism. This paired design enables precise identification of genomic differences associated with the altered phenotype. . Both long and short-read technologies are suitable for the sequencing of microbial genome analysis. General recommendations for sequencing technologies and indicative sequencing depths for microbial studies are summarized in Table 6. Following genome sequencing and annotation, priority should be given to the analysis of genomic regions associated with regulatory functions, secondary metabolite biosynthesis, and stress response pathways. A suggested workflow for sequencing-based analysis of beneficial microorganisms is shown in **Figure 4**.

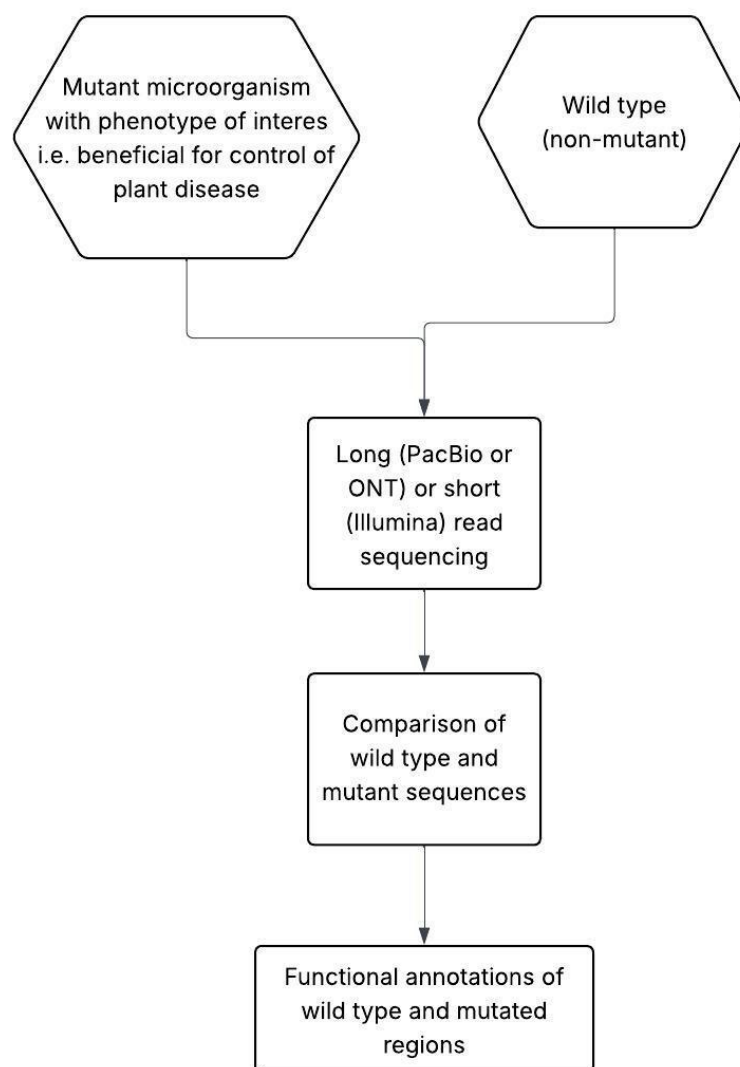


Figure 4. Suggested workflow for sequencing of beneficial microorganisms

Table 6. Recommended target depth of sequencing by technology for microorganisms

Sequencing technology	Target depth	Best use
PacBio HiFi	20-30X	Assembly, structural variants (SV), SNPs, Indels discovery
ONT	50-80X	Assembly and SV discovery
	80-100x	SNPs, Indels discovery
Illumina	100-200X(*)	SNPs, Indels discovery
ONT + Illumina	40-60X ONT, 100X Illumina	Assembly, structural variants (SV), SNPs, Indels discovery

(*) suggested for mixed heterogenous populations of bacteria or fungus

IV. Post-sequencing phase

The post-sequencing phase includes retrieval of raw sequencing data, assessment of data quality, genome assembly or read mapping, genome annotation, and downstream analyses aimed at identifying genomic regions associated with mutant phenotypes. This section provides a conceptual overview only; detailed bioinformatic workflows and tool-specific protocols will be developed in Products 2 and 3.

1. Evaluate the quality of the sequencing data

This step is critical in determining whether downstream analysis can be reliably performed. Quality control of sequencing reads typically includes assessment of base-call accuracy (Q-scores), nucleotide composition (e.g. GC content), presence of technical artifacts (e.g. adapters and barcodes), and consistency between read pairs for paired-end data. Commonly used tools for quality assessment of the data (reads) using conventional tools such as fastQC for short read data (Andrews, 2010), NanoPlot and NanoFilt (De Coster, et al., 2018) and longQC (Fukasawa et al., 2020) for long read data. These tools provide diagnostic metrics that allow users to determine whether the data meets minimum quality requirements for further analysis.

2. Remove sequencing artifacts

Identified artifacts including adapters, barcodes, and low-quality bases, should be removed prior to genome assembly or read mapping. This step reduces biases introduced during library preparation and improves the accuracy of downstream analyses. Several tools are available to address this, such as fastp (Chen et al., 2018; Chen, 2023; Chen 2025) and cutadapt (Martin, 2011) among others.

3. Reads mapping or de novo assembly

The choice between read mapping and de novo genome assembly depends on the availability and suitability of a reference genome and on the sequencing technology used. When a high-quality reference genome is available, sequencing reads can be mapped directly to this reference to identify genetic variants between mutant and control samples. This approach is generally appropriate for resequencing studies aimed at detecting single-nucleotide variants, small insertions or deletions, and copy number changes.

In the absence of a suitable reference genome, or when major structural rearrangements are expected, de novo genome assembly may be required. Long-read sequencing technologies are particularly well suited for de novo assembly, whereas short-read technologies are typically used for reference-based sequencing. De novo assemblies for short read data include SPAdes (Bankevich et al., 2012) and SOAPdenovo2 (Luo et al., 2012) among others. For mapping short reads to a reference genome, BWA-MEM (Li, 2013), and Bowtie2 (Langmead and Salzberg, 2012) among others. For de novo assembly of long sequencing Flye (Kolmogorov et al., 2019) for PacBio and ONT reads; and Hifiasm (Cheng et al., 2018) for PacBio reads. For mapping long reads to a reference genome, Minimap2 (Li, 2018) for PacBio and ONT reads; and NGMLR (Sedlazeck et al., 2018) optimized for structural variant detection.

4. Evaluate the genome assembly quality

Evaluations of genome assembly rely on three key parameters; continuity, completeness and correctness and are known as the 3C criterium.

Contiguity or continuity - continuity reflects how fragmented or contiguous an assembly is, it indicates for how long the reads are being assembled on a given contig, the longer the N50 the better the continuity. The auN metric has also been used and is commonly reported by genome quality assessment tools such as GAGE (Salzberg et al., 2012) and QUAST (Gurevich et al., 2013). The CC ratio of Contig/Chromosome has been suggested as a more accurate substitute to the N50, it represents the ratio of contig counts to the chromosome pair number of an assembled genome. The interpretation is straight forward, the more continuous the assemble the smaller the CC ratio (Wang and Wang, 2023). **Completeness** - completeness refers to the proportion of the genome and gene space represented in the assembly. It can be measured using tools like K-mer analysis by comparing the total k-mer numbers in the reads against the expected copy number (Formenti et al., 2022). To assess completeness of the gene space, BUSCO (Manni et al., 2021), PLAZA CoreGF (Van Bel et al., 2012), CEGMA (Parra et al., 2007) are suggested tools. CheckM (Parks et al., 2015) is a very specific tool for evaluating completeness of prokaryotic genomes but also can work with fungal genomes. **Correctness** - correctness refers to the structural and base-level accuracy of the assembly but it is difficult to be measured. If a reference genome is available, the metric can be extracted from the degree of similarity and the identification of high confidence regions in the re-sequenced genome. Alternatively the base-level error rate and structural errors can be determine using existing tools such as K-mer analysis (Koren et al., 2018; Chen et al., 2021). A summary of the recommended parameters and expected thresholds for evaluating continuity, completeness and correctness is present in Wang and Wang, 2023 and is being included in **Table 7**.

Table 7. Metrics for evaluating the quality of a genome assembly (Taken from Wang and Wang, 2023)

Dimension	Metric	Score for a finished assembly
Contiguity	N50	Chromosome N50
	CC ratio ^a	1
Completeness	Overall completeness: k-mer-based	100%
	Gene space completeness: BUSCO ^b	Near 100% ^c
	Tandem repeat completeness: telomeric and sub-telomeric satellite arrays, centromeric satellite arrays, ribosomal DNA loci	100%
Correctness	Base-level error rate	0%
	Structural error: collapse, inversion, false duplication, chimeric joins	0%

^a Ratio of contig counting to chromosome pair number

^b Benchmarking universal single-copy orthologs

^c BUSCO value should tolerate a range for species in specialized habitat

5. Genome annotation

Genome annotation involves two main components: **structural annotation**, which identifies gene boundaries and genomic features (exons, introns, untranslated regions (UTRs) and the CDS out of translated regions) and **functional annotation**, which assigns putative functions to predicted genes based on similarity to known sequences and biological pathways. Structural annotation is typically performed after genome assembly, while functional annotation may be tailored to the biological

question under investigation. One of the most commonly used annotators is Augustus (Stanke et al., 2005) for annotating eukaryotic genomes including plant and fungi. Augustus is an “ab initio” program meaning it based its predictions uniquely on the sequence. More sophisticated approaches like BRAKER2 (Brüna et al., 2021) for annotating plant genomes, also use the prediction capabilities from Augustus and integrating it with protein homology evidence. Closely related genomes that are fully annotated can also be used to do annotation transfers using tools like TOGA (Kirilenko et al., 2023). There are several tools for annotating bacterial genomes, some of them are web based applications such as BASys2 (Van Domselaar et al, 2005; Polzer et al., 2025) or command line tools such as PGAP (Tatusova et al., 2016; Li et al., 2021) among others.

6. Downstream analysis of the genome(s)

Downstream analyses depend on the phenotype of interest and the experimental design. For example, when a non-mutant control (WT) is available, variant calling approaches can be used to identify genetic differences between mutant and control genotypes. Structural variant analysis may be required when large insertions, deletions, or rearrangements are suspected. When only a reference genome is available but not a non-mutant genotype, analyses may focus on functionally relevant genomic regions, such as disease resistance genes (or genes involved in key metabolic pathways). Regardless of the analytical strategy, candidate genomic changes identified through bioinformatic analysis should be validated experimentally whenever possible.

Specific bioinformatic workflows using a specific tool for each step of the post-sequencing part will be developed in Product II.

V. References

- Adero, M., Tripathi, J.N. & Tripathi, L. 2023. Advances in somatic embryogenesis of banana. *International Journal of Molecular Sciences*, 24: 10999.
- Arshad, R., Razzaq, T., Ahmad, B., Hou, T., Li, C., Jin, Z., Zhang, W., Liu, Z., Huang, H.-R., Lü, P. & et al. 2025. Banana breeding by genome design. *Journal of Integrative Plant Biology*, 67: 2816–2847.
- Ahloowalia, B.S., Maluszynski, M. & Nichterlein, K. 2004. Global impact of mutation-derived varieties. *Euphytica*, 135: 187–204.
- Akopyan, M., Genchev, M., Armstrong, E.E. & Mooney, J.A. 2025. Reference genome choice compromises population genetic analyses. *Cell*, 188(24): 6939–6952.e11. <https://doi.org/10.1016/j.cell.2025.08.034>
- Bankevich, A., Nurk, S., Antipov, D. & et al. 2012. SPAdes: A new genome assembly algorithm and its applications to single-cell sequencing. *Journal of Computational Biology*, 19: 455–477. <https://doi.org/10.1089/cmb.2012.0021>
- Belser, C., Istace, B., Denis, E., Dubarry, M., Baurens, F.C., Falentin, C., Genete, M., Berrabah, W., Chèvre, A.M., Delourme, R. & et al. 2018. Chromosome-scale assemblies of plant genomes using nanopore long reads and optical maps. *Nature Plants*, 4: 879–887.
- Belser, C., Baurens, F.C., Noel, B., Martin, G., Cruaud, C., Istace, B., Yahiaoui, N., Labadie, K., Hřibová, E., Doležel, J. & et al. 2021. Telomere-to-telomere gapless chromosomes of banana using nanopore sequencing. *Communications Biology*, 4: 1047.
- Bentley, D.R., Balasubramanian, S., Swerdlow, H.P., Smith, G.P., Milton, J., Brown, C.G., Hall, K.P., Evers, D.J., Barnes, C.L., Bignell, H.R., Boutell, J.M., Bryant, J., Carter, R.J., Keira Cheetham, R., Cox, A.J., Ellis, D.J., Flatbush, M.R. & et al. 2008. Accurate whole human genome sequencing using reversible terminator chemistry. *Nature*, 456: 53–59. <https://doi.org/10.1038/nature07517>
- Brúna, T., Hoff, K.J., Lomsadze, A., Stanke, M. & Borodovsky, M. 2021. BRAKER2: Automatic Eukaryotic Genome Annotation with GeneMark-EP+ and AUGUSTUS Supported by a Protein Database. *NAR Genomics and Bioinformatics*. <https://doi.org/10.1093/nargab/lqaa108>
- Chen, N.C., Solomon, B., Mun, T., Iyer, S. & Langmead, B. 2021. Reference flow: Reducing reference bias using multiple population genomes. *Genome Biology*, 22(1): 8. <https://doi.org/10.1186/s13059-020-02229-3>
- Chen, Y. & et al. 2021. Accurate long-read de novo assembly evaluation with Inspector. *Genome Biology*, 22: 312.
- Chen, S. 2025. fastp 1.0: An ultra-fast all-round tool for FASTQ data quality control and preprocessing. *iMeta*, 2025. <https://doi.org/10.1002/imt2.107>
- Chen, S. 2023. Ultrafast one-pass FASTQ data preprocessing, quality control, and deduplication using fastp. *iMeta*, 2: e107. <https://doi.org/10.1002/imt2.107>
- Chen, S., Zhou, Y., Chen, Y. & Gu, J. 2018. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17): i884–i890. <https://doi.org/10.1093/bioinformatics/bty560>
- Cheng, H., Concepcion, G.T., Feng, X., Zhang, H. & Li, H. 2021. Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm. *Nature Methods*, 18: 170–175. <https://doi.org/10.1038/s41592-020-01056-5>
- Datta, S., Jankowicz-Cieslak, J., Nielen, S. & et al. 2018. Induction and recovery of copy number variation in banana through gamma irradiation and low-coverage whole-genome sequencing. *Plant Biotechnology Journal*, 16: 1644–1653.

- Deamer, D., Akeson, M. & Branton, D. 2016. Three decades of nanopore sequencing. *Nature Biotechnology*, 34: 518–524. <https://doi.org/10.1038/nbt.3423>
- Doyle, J.J. & Doyle, J.L. 1987. A rapid DNA isolation procedure for small quantities of fresh leaf tissue. *Phytochemical Bulletin*, 19: 11–15.
- Eid, J., Fehr, A., Gray, J., Luong, K., Lyle, J., Otto, G., Peluso, P., Rank, D., Baybayan, P., Bettman, B., Bibillo, A., Bjornson, K., Chaudhuri, B., Christians, F., Cicero, R., Clark, S. & et al. 2009. Real-time DNA sequencing from single polymerase molecules. *Science*, 323: 133–138. <https://doi.org/10.1126/science.1162986>
- Formenti, G. & et al. 2022. Merfin: Improved variant filtering, assembly evaluation and polishing via k-mer validation. *Nature Methods*, 19: 696–704.
- Gand, M., Bloemen, B., Vanneste, K., Roosens, N.H.C. & De Keersmaecker, S.C.J. 2023. Comparison of 6 DNA extraction methods for isolation of high yield of high molecular weight DNA suitable for shotgun metagenomics Nanopore sequencing to detect bacteria. *BMC Genomics*, 24(1): 438. <https://doi.org/10.1186/s12864-023-09537-5>
- Goodwin, S., McPherson, J.D. & McCombie, W.R. 2016. Coming of age: ten years of next-generation sequencing technologies. *Nature Reviews Genetics*, 17: 333–351. <https://doi.org/10.1038/nrg.2016.49>
- Healey, A., Furtado, A., Cooper, T. & Henry, R.J. 2014. Protocol: a simple method for extracting next-generation sequencing quality genomic DNA from recalcitrant plant species. *Plant Methods*, 10: 21. <https://doi.org/10.1186/1746-4811-10-21>
- Huang, H.-R., Liu, X., Arshad, R., Wang, X., Li, W.-M., Zhou, Y. & Ge, X.J. 2023. Telomere-to-telomere haplotype-resolved reference genome reveals subgenome divergence and disease resistance in triploid Cavendish banana. *Horticulture Research*, 10: uhad153.
- Hwang, S.C. & Ko, W.H. 2007. Cavendish banana cultivars resistant to fusarium wilt acquired through somaclonal variation in Taiwan. *Plant Disease*, 88: 580–588.
- Jain, C., Rhie, A., Zhang, H. & et al. 2020. Weighted minimizer sampling improves long-read mapping. *Bioinformatics*, 36: i111–i118. <https://doi.org/10.1093/bioinformatics/btaa435>
- Jain, M., Fiddes, I., Miga, K. & et al. 2015. Improved data analysis for the MinION nanopore sequencer. *Nature Methods*, 12: 351–356. <https://doi.org/10.1038/nmeth.3290>
- Jankowicz-Cieslak, J., Huynh, O.A., Brozynska, M. & et al. 2012. Induction, rapid fixation and retention of mutations in vegetatively propagated banana. *Plant Biotechnology Journal*, 10: 1056–1066.
- Kirilenko, B.M., Munegowda, C., Osipova, E. & et al. 2023. Integrating gene annotation with orthology inference at scale. *Science*, 380(6643): eabn3107.
- Koh, R.B.L., Barbosa, C.F.C., Aquino, V.M. & Galvez, L.C. 2021. Extraction of high molecular weight DNA suitable for next-generation sequencing from the fiber crop abaca. *Industrial Crops and Products*, 161: 113194. <https://doi.org/10.1016/j.indcrop.2020.113194>
- Kolmogorov, M., Yuan, J., Lin, Y. & Pevzner, P.A. 2019. Assembly of long, error-prone reads using repeat graphs. *Nature Biotechnology*, 37: 540–546. <https://doi.org/10.1038/s41587-019-0072-8>
- Koren, S. & et al. 2018. De novo assembly of haplotype-resolved genomes with trio binning. *Nature Biotechnology*, 36: 1174–1182.
- Koren, S., Walenz, B.P., Berlin, K. & et al. 2017. Canu: scalable and accurate long-read assembly via adaptive k-mer weighting and repeat separation. *Genome Research*, 27: 722–736. <https://doi.org/10.1101/gr.215087.116>

- Langmead, B. & Salzberg, S.L. 2012. Fast gapped-read alignment with Bowtie 2. *Nature Methods*, 9: 357–359. <https://doi.org/10.1038/nmeth.1923>
- Li, H. 2013. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv*: 1303.3997. <https://arxiv.org/abs/1303.3997>
- Li, H. 2018. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34: 3094–3100. <https://doi.org/10.1093/bioinformatics/bty191>
- Li, W., O'Neill, K., Haft, D.H. & et al. 2021. RefSeq: expanding the Prokaryotic Genome Annotation Pipeline reach with protein family model curation. *Nucleic Acids Research*, 49(D1): D1020–D1028. <https://doi.org/10.1093/nar/gkaa1005>
- Li, Z., Wang, J., Fu, Y., Jing, Y., Huang, B., Chen, Y., Wang, Q., Wang, X.B., Meng, C., Yang, Q. & et al. 2022. The *Musa troglodytarum* L. genome provides insights into the mechanism of non-climacteric behavior and enrichment of carotenoids. *BMC Biology*, 20: 186.
- Li, X., Yu, S., Cheng, Z., Chang, X., Yun, Y., Jiang, M., Chen, X., Wen, X., Li, H., Zhu, W. & et al. 2024. Origin and evolution of the triploid cultivated banana genome. *Nature Genetics*, 56: 136–142.
- Liu, X., Arshad, R., Wang, X., Li, W.M., Zhou, Y., Ge, X.J. & Huang, H.R. 2023. The phased telomere-to-telomere reference genome of *Musa acuminata*, a main contributor to banana cultivars. *Scientific Data*, 10: 631.
- Luo, R., Liu, B., Xie, Y. & et al. 2012. SOAPdenovo2: an empirically improved memory-efficient short-read de novo assembler. *GigaScience*, 1: 18. <https://doi.org/10.1186/2047-217X-1-18>
- Lv, N., Tao, C., Ou, Y., Wang, J., Deng, X., Liu, H. & et al. 2023. Root-associated antagonistic *Pseudomonas* spp. contribute to soil suppressiveness against banana *Fusarium* wilt disease. *Microbiology Spectrum*, 11: e0352522. <https://doi.org/10.1128/spectrum.03525-22>
- Martin, G., Istace, B., Baurens, F.C., Belser, C., Hervouet, C., Labadie, K., Cruaud, C., Noel, B., Guionou, C., Salmon, F. & et al. 2025. Unravelling genomic drivers of speciation in *Musa* through genome assemblies of wild banana ancestors. *Nature Communications*, 16: 1–4.
- Mintoff, S.J.L., Nguyen, T.V., Kelly, C., Cullen, S., Hearnden, M., Williams, R., Daniels, J.W. & Tran-Nguyen, L.T.T. 2021. Banana cultivar field screening for resistance to *Fusarium oxysporum* f. sp. *cubense* tropical race 4 in the Northern Territory. *Journal of Fungi*, 7(8): 627. <https://doi.org/10.3390/jof7080627>
- Palmieri, D., Ianiri, G., Del Grosso, C., Barone, G., De Curtis, F., Castoria, R. & Lima, G. 2022. Advances and perspectives in the use of biocontrol agents against fungal plant diseases. *Horticulturae*, 8(7): 577. <https://doi.org/10.3390/horticulturae8070577>
- Papoutsoglou, E.A., Filleul, E., Frainay, C., Baker, C., Holub, P., Gandon, F., Poorter, H., Kattge, J., Devi, S., van Rijswijk, J., Bonnett, D., Arnaud, E., Kersey, P.J., Kreil, D.P., Lange, M., Laporte, M.-A., Michotey, C., Oppermann, M., Michotey, P. & et al. 2020. Enabling reusability of plant phenotypic datasets with MIAPPE 1.1. *New Phytologist*, 227(1): 260–273. <https://doi.org/10.1111/nph.16544>
- Parra, G., Bradnam, K. & Korf, I. 2007. CEGMA: A pipeline to accurately annotate core genes in eukaryotic genomes. *Bioinformatics*, 23: 1061–1067.
- Rostami, M., Ghorbani, A. & Shahbazi, S. 2024. Gamma radiation-induced enhancement of biocontrol agents for plant disease management. *Current Research in Microbial Sciences*, 7: 100308. <https://doi.org/10.1016/j.crmicr.2024.100308>

- Sedlazeck, F.J., Rescheneder, P., Smolka, M. & et al. 2018. Accurate detection of complex structural variations using single-molecule sequencing. *Nature Methods*, 15: 461–468.
<https://doi.org/10.1038/s41592-018-0001-7>
- Shan, W., Liu, H., Zhou, Y. & Yu, X. 2017. Draft genome sequence of *Streptomyces* sp. XY006, an endophyte isolated from tea (*Camellia sinensis*). *Genome Announcements*, 5: 00971–00917.
<https://doi.org/10.1128/genomeA.00971-17>
- Shan, W., Zhou, Y., Liu, H. & Yu, X. 2018. Endophytic actinomycetes from tea plants (*Camellia sinensis*): Isolation, abundance, antimicrobial, and plant-growth-promoting activities. *BioMed Research International*, 2018: 1–12. <https://doi.org/10.1155/2018/1470305>
- Stanke, M. & Morgenstern, B. 2005. AUGUSTUS: a web server for gene prediction in eukaryotes that allows user-defined constraints. *Nucleic Acids Research*, 33(14): W465–W467.
<https://doi.org/10.1093/nar/gki462>
- Tao, C., Wang, Z., Liu, S., Lv, N., Deng, X., Xiong, W. & et al. 2023. Additive fungal interactions drive biocontrol of *Fusarium* wilt disease. *New Phytologist*, 238: 1198–1214.
<https://doi.org/10.1111/nph.18793>
- Tatusova, T., DiCuccio, M., Badretdin, A. & et al. 2016. NCBI prokaryotic genome annotation pipeline. *Nucleic Acids Research*, 44(14): 6614–6624. <https://doi.org/10.1093/nar/gkw569>
- Van Bel, M., Proost, S., Wischnitzki, E., Movahedi, S., Scheerlinck, C., Van de Peer, Y. & Vandepoele, K. 2012. Dissecting plant genomes with the PLAZA comparative genomics platform. *Plant Physiology*, 158: 590–600.
- Van Domselaar, G.H., Stothard, P., Shrivastava, S., Cruz, J.A., Guo, A., Dong, X., Lu, P., Szafron, D., Greiner, R. & Wishart, D.S. 2005. BASys: a web server for automated bacterial genome annotation. *Nucleic Acids Research*, 33: W455–W459.
- Wang, Z., Miao, H., Liu, J., Xu, B., Yao, X., Xu, C., Zhao, S., Fang, X., Jia, C., Wang, J. & et al. 2019. *Musa balbisiana* genome reveals subgenome evolution and functional divergence. *Nature Plants*, 5: 810–821.
- Wang, W., Niu, L. & Hou, C. 2022. Interrogating global chromatin interaction network by high-throughput chromosome conformation capture (Hi-C) in plants. *Methods in Molecular Biology*, 2484: 55–67. https://doi.org/10.1007/978-1-0716-2253-7_5
- Wenger, A.M., Peluso, P., Rowell, W.J., Chang, P.-C., Hall, R.J. & et al. 2019. Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome. *Nature Biotechnology*, 37: 1155–1162. <https://doi.org/10.1038/s41587-019-0217-9>
- Wenger, A.M., Peluso, P., Rowell, W.J., Chang, P.-C., Hall, R.J., Concepcion, G.T., Ebler, J., Fungtammasan, A., Kolesnikov, A., Olson, N.D., Töpfer, A., Alonge, M., Mahmoud, M., Qian, Y., Chin, C.-S., Phillippy, A.M., Schatz, M.C., Myers, G., DePristo, M.A. & Hunkapiller, M.W. 2019. Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome. *Nature Biotechnology*, 37: 1155–1162. <https://doi.org/10.1038/s41587-019-0217-9>
- Xiang, D., Yang, X., Liu, B., Chu, Y., Liu, S. & Li, C. 2023. Bio-priming of banana tissue culture plantlets with endophytic *Bacillus velezensis* EB1 to improve *Fusarium* wilt resistance. *Frontiers in Microbiology*, 14: 1146331. <https://doi.org/10.3389/fmicb.2023.1146331>
- Xiao, T.W., Liu, X., Fu, N., Liu, T.J., Wang, Z.F., Ge, X.J. & Huang, H.R. 2024. Chromosome-level genome assemblies of *Musa ornata* and *M. velutina* provide insights into pericarp dehiscence and anthocyanin biosynthesis in banana. *Horticulture Research*, 11: uhae079.

- Zhou, D., Jing, T., Qi, D., Renjun, F., Yajie, D., Yufeng, C. & et al. 2017. Isolation and identification of *Streptomyces lunalinharesii* and its control effect on banana Fusarium wilt disease. *Acta Horticulturae Sinica*, 44: 664–674. <https://doi.org/10.16420/j.issn.0513-353x.2016-0598>
- Zhou, R., Wang, S., Zhan, N., He, W., Deng, G., Dou, T., Zhu, X.-T., Xie, W.-Z., Zheng, Y.-Y., Hu, C., Bi, F., Gao, H., Dong, T., Liu, S., Li, C., Yang, Q., Wang, L., Song, J.-M., Dang, J., Guo, Q., Yi, G., Chen, L.-L. & Sheng, O. 2024. High-quality genome assemblies for two *Australimusa* bananas (*Musa* spp.) and insights into regulatory mechanisms of superior fiber properties. *Plant Communications*, 5: 100681. <https://doi.org/10.1016/j.xplc.2023.100681>